

Maximising Revenue for Drivers



Problem Statement

The taxi booking industry relies on data-driven decisions to maximise driver earnings and improve operational efficiency. Using a dataset of 6.4+ million records, this project analyses the relationship between payment methods and fare amounts to identify patterns that influence revenue. The insights help understand whether payment type affects fare pricing and support strategies for maximising driver revenue.

Research Question

To what extent does the payment method impact the total fare collected?

Can strategic payment preferences be promoted to enhance driver revenue without affecting customer satisfaction?

Data Overview

For this analysis, we utilised the comprehensive NYC Taxi Trip dataset. We performed data cleaning and feature engineering to prepare the data, retaining only the relevant columns required for the analysis. The dataset consisted of 18 original columns and one derived feature, *Duration*, calculated as the difference between **tpep_dropoff_datetime** and **tpep_pickup_datetime**.

Relevant columns used for this research:

passenger_count (1 to 5)

payment_type (card or cash)

fare_amount

trip_distance (miles)

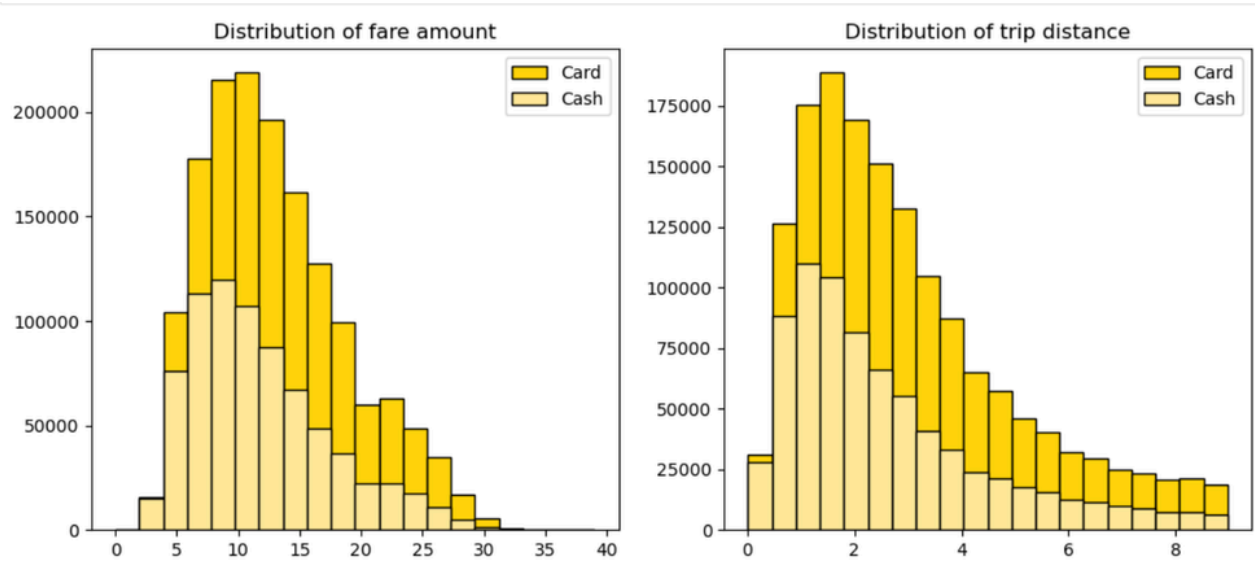
duration (minutes)

	passenger_count	payment_type	fare_amount	trip_distance	duration
0	1.0	1.0	6.00	1.20	4.800000
1	1.0	1.0	7.00	1.20	7.416667
2	1.0	1.0	6.00	0.60	6.183333
3	1.0	1.0	5.50	0.80	4.850000
4	1.0	2.0	3.50	0.00	2.300000
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Methodology

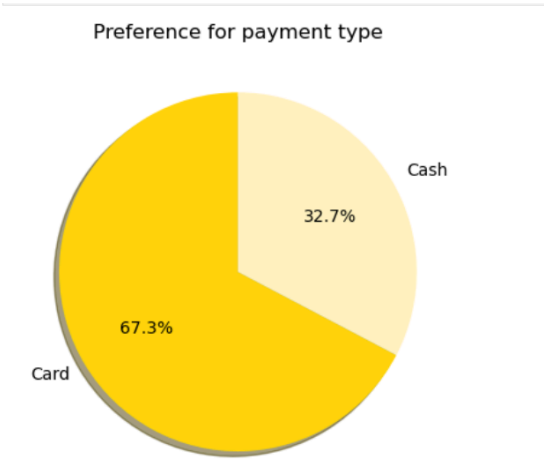
Step	Description
Descriptive Analysis	Performed exploratory and descriptive statistical analysis to summarize the dataset, focusing on key metrics such as fare amount , payment type , and trip duration .
Hypothesis Testing	Conducted an independent two-sample T-test to determine whether a statistically significant relationship exists between payment type and fare amount , testing whether different payment methods are associated with different average fares.

Journey Insights



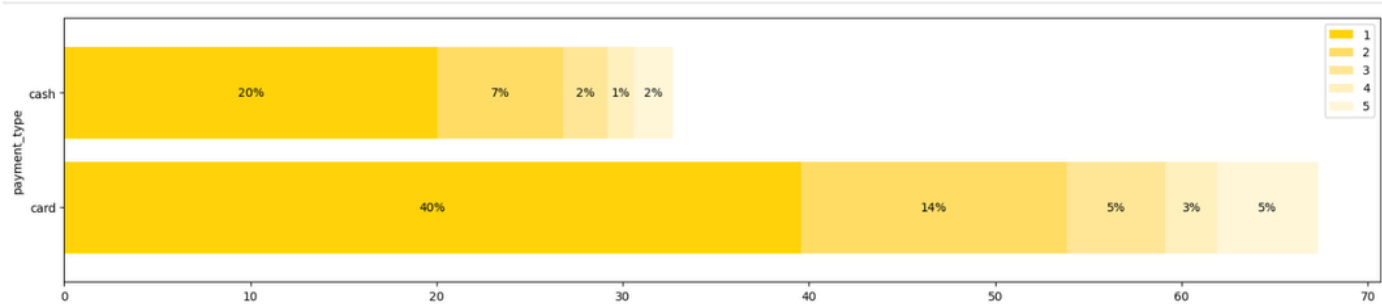
- Customers paying by card have a slightly higher average trip distance and fare amount than those paying by cash, indicating greater revenue potential from digital payment users.
- The analysis suggests that customers are more likely to choose card payments for higher-value and longer-distance trips, making digital payments a key driver of increased revenue.

Preference of Payment Types



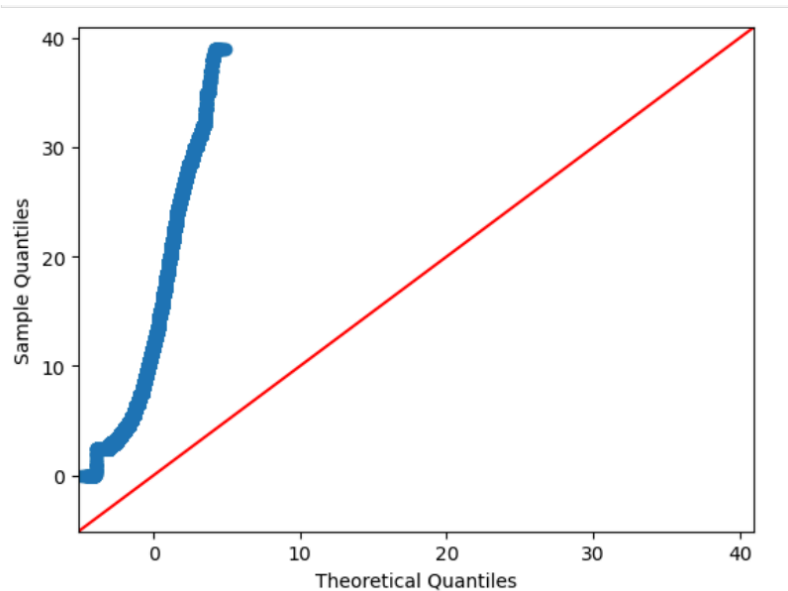
- Card payments dominate customer transactions, accounting for 67.5% of all trips, while cash payments represent only 32.5%, highlighting a strong customer preference for digital payments.
- The high adoption of card payments suggests that customers value the convenience, security, and seamless experience of cashless transactions, making digital payments an effective channel for increasing driver revenue.

Passenger Count Analysis



- Single-passenger rides dominate taxi usage, with 40.08% of all card transactions and 20.04% of all cash transactions involving only one passenger, making solo travellers the primary customer segment.
- The proportion of trips decreases as passenger count increases, indicating that larger groups are less likely to use taxis and may prefer alternative modes of transportation.
- Understanding the relationship between payment method and passenger count enables targeted strategies, such as personalised offers for frequent solo travellers and optimised pricing to maximise driver revenue.

Hypothesis Testing



- The hypothesis test confirms a statistically significant difference in average fares between card and cash payments (T-statistic = 169.21, P-value < 0.05), indicating that payment method is associated with fare amount.
- Since the null hypothesis is rejected, payment type is an important factor in understanding customer spending behaviour and should be considered in revenue optimisation strategies.
- Encouraging digital (card) payments can help increase overall driver earnings, as card transactions are consistently associated with higher average fares.

Recommendations

- Encourage customers to use credit cards for taxi payments, as card transactions are associated with higher average fares and longer trips, creating greater revenue opportunities for drivers while providing a more convenient payment experience.
- Introduce incentives for digital payments, such as cashback, discounts, or loyalty rewards, to encourage customers to choose card payments for their trips.
- Enhance the digital payment experience by providing fast, secure, and seamless card payment options, increasing customer satisfaction while driving higher revenue for drivers.